

Adapting the Harrow-Hassidim-Lloyd algorithm to quantum many-body theory

Paper Review

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Problem Setup

Coupled Cluster Ansatz:

$$|\Psi\rangle = e^T |\Phi_0\rangle$$

- $|\Phi_0\rangle$: Hartree–Fock (reference state)
- $T = \sum_Q t_Q \hat{e}_Q$: cluster operator (excitations)
- t_Q : cluster amplitudes

LCC approximation:

$$e^T \approx 1 + T \quad \Rightarrow \quad |\Psi\rangle \approx (1 + T)|\Phi_0\rangle$$

Projected Schrödinger equation:

$$\langle \chi_P | H | \Psi \rangle = 0$$

$$\langle \chi_P | H | \Phi_0 \rangle + \sum_Q \langle \chi_P | H \hat{e}_Q | \Phi_0 \rangle t_Q = 0$$

Rewriting:

$$\sum_Q \langle \chi_P | H | \chi_Q \rangle t_Q = -\langle \chi_P | H | \Phi_0 \rangle$$

$$\Rightarrow \quad A t = -b \implies t = -A^{-1} b$$

Correlation Energy and Measurement

Correlation energy:

Energy correction beyond the Hartree–Fock approximation due to electron correlation.

From LCC equations:

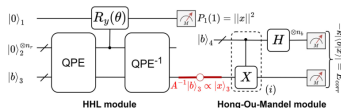
$$E_{\text{corr}} = \langle \Phi_0 | H | \Psi \rangle - E_{\text{HF}}$$
$$E_{\text{corr}} = \sum_Q b_Q t_Q = b^T t = b^T A^{-1} b$$

Quantum interpretation:

$$|x\rangle \propto A^{-1}|b\rangle \quad \Rightarrow \quad E_{\text{corr}} \propto \langle b|x\rangle$$

Measurement strategy:

- Prepare quantum states $|b\rangle$ and $|x\rangle$
- Interfere them using a Hong–Ou–Mandel (HOM) circuit



HOM circuit used to measure overlap

HHL Algorithm: Solving $Ax = b$

Goal

$$|x\rangle \propto A^{-1}|b\rangle$$

Eigenbasis Representation

$$A|u_j\rangle = \lambda_j|u_j\rangle \implies |b\rangle = \sum_j \beta_j|u_j\rangle$$

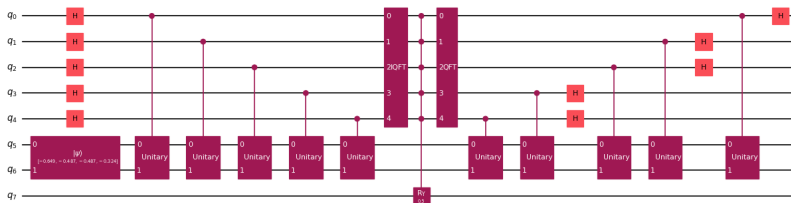
Core Idea

- Use QPE to estimate eigenvalues λ_j
- Apply controlled rotation $\sim 1/\lambda_j$
- Uncompute to obtain solution state

Result

$$|x\rangle \propto \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

HHL Circuit Structure



HHL circuit showing QPE, controlled rotation, and uncomputation

Registers:

- **Clock register:** encodes eigenvalues λ_j via QPE
- **System register:** stores input $|b\rangle$ and evolves to $|x\rangle$
- **Ancilla qubit:** performs controlled rotation $\sim 1/\lambda_j$

Results: H₂ System

System: Matrix A and vector b obtained from LCC equations for the H₂ molecule.

Matrix A :

$$\begin{bmatrix} 1.12854 & 0 & 0 & 0.03593 \\ 0 & 1.44616 & 0.08368 & 0 \\ 0 & 0.08368 & 1.44616 & 0 \\ 0.03593 & 0 & 0 & 1.94607 \end{bmatrix}$$

Vector b :

$$[-0.2, -0.15, -0.15, -0.1]$$

Properties:

- Symmetric matrix
- Eigenvalues: $\sim [1.13, 1.36, 1.53, 1.95]$
- Condition number: $\kappa \approx 1.73$

Classical solution:

$$x = -A^{-1}b$$

$$E_{\text{corr}} = -0.06937$$

Quantum (HHL):

$$x_{\text{quantum}} \sim [0.246, 0.678, 0.662, 0.207]$$

$$E_{\text{corr}} \approx -0.06198$$

Observation:

- Good qualitative agreement
- Noticeable error in E_{corr} ($\approx 10.6\%$)

Limitations of HHL

Controlled unitaries:

$$U^{2^k} = e^{iAt2^k} \Rightarrow \text{large circuit depth}$$

QPE precision tradeoff:

$$\lambda_j \sim \frac{k}{2^{n_r}} \Rightarrow \text{precision vs resources}$$

Rotation instability:

$$\theta \sim \frac{1}{\lambda_j} \Rightarrow \text{errors for small } \lambda_j$$

Error propagation:

$$\delta\lambda_j \Rightarrow \delta\left(\frac{1}{\lambda_j}\right) \Rightarrow |x\rangle$$

Result: $\sim 10\%$ deviation in E_{corr}

To mitigate this, we introduce:

HHLite and AdaptHHLite

HHLite(implementation on truncated eigenspace): Core Idea

Motivation: Full HHL requires precise estimation of all eigenvalue bits using QPE.

Key observation:

$$\lambda_j = 0.b_1b_2 \dots b_{n_r}$$

Not all bits contribute equally to the final result.

HHLite idea:

Fix dominant eigenvalue bits and ignore the rest

Using QPE measurements:

- Measure the clock qubits to extract **dominant outcomes**
- Probability that qubit i is measured as 1: $P_i(1)$

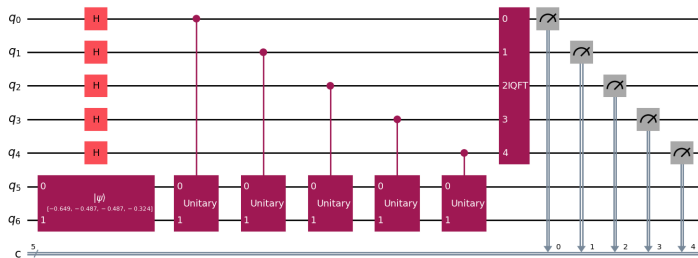
Introduce threshold:

$$p_{\text{th}} \approx 0.8$$

$$P_i(1) > 0.8 \Rightarrow \text{fix qubit to } |1\rangle, P_i(1) < 0.2 \Rightarrow \text{fix qubit to } |0\rangle$$

Result: Fixed qubits remove control — U^{2^k} applied directly for 1, skipped for 0, reducing circuit depth

QPE Output and Multiqubit Fixing (H_2)



QPE circuit used for the H_2 system

Per-qubit probabilities (H_2): $P_i(1) = [0.831, 0.602, 0.857, 0.395, 0.511]$

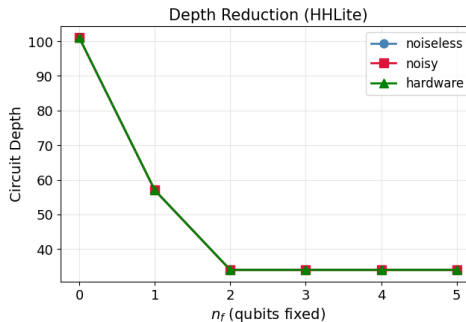
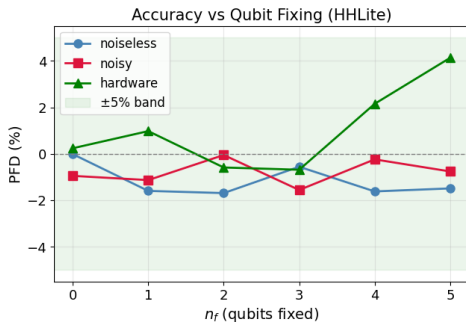
Fixing priority (by confidence): $[2, 0, 3, 1, 4]$

Interpretation:

- Qubits 2 and 0 are strongly biased $\Rightarrow$ fixed first
- Qubits near 0.5 remain unfixed

Simulation Results: HHLite (H_2)

HHLite: Noiseless vs Noisy vs Hardware



$$\text{PFD} = \frac{E_{\text{quantum}} - E_{\text{classical}}}{|E_{\text{classical}}|} \times 100$$

AdaptHHLite: Matrix Scaling

Key idea: Scale the matrix A so that its eigenvalues are easy to represent and invert.

Define:

$d_{\min}, d_{\max}$ = smallest and largest diagonal elements of A

Scaling:

$$A \rightarrow \tilde{A} = sA, \quad s = \frac{2^{-n_r}}{d_{\min}}$$

- Ensures eigenvalues of $\tilde{A}$ lie in a controlled range
- Represents smallest eigenvalue in n_r bits

Controlled rotation:

$$\theta_i = 2 \arcsin\left(\frac{c}{\tilde{\lambda}_i}\right), \text{ choose } c = 2^{-n_r}$$

$$\tilde{\lambda}_i = \frac{i}{2^{n_r}} \Rightarrow \frac{c}{\tilde{\lambda}_i} = \frac{1}{i}$$

Result: Simpler and fixed controlled rotations $\Rightarrow$ reduced circuit complexity. Combining this with the previous **Lite Module** forms **AdaptHHLite**.

AdaptHHLite on H_2 : Matrix Scaling

Original matrix A :

$$\begin{bmatrix} 1.1285 & 0 & 0 & 0.0359 \\ 0 & 1.4462 & 0.0837 & 0 \\ 0 & 0.0837 & 1.4462 & 0 \\ 0.0359 & 0 & 0 & 1.9461 \end{bmatrix}$$

Scaling parameters:

$$d_{\min} = 1.12854, \quad s = 0.02769, \quad c = 2^{-nr} = 0.03125$$

Scaled matrix $\tilde{A} = sA$:

$$\begin{bmatrix} 0.03125 & 0 & 0 & 0.00100 \\ 0 & 0.04003 & 0.00232 & 0 \\ 0 & 0.00232 & 0.04003 & 0 \\ 0.00100 & 0 & 0 & 0.05393 \end{bmatrix}$$

Scaled eigenvalues:

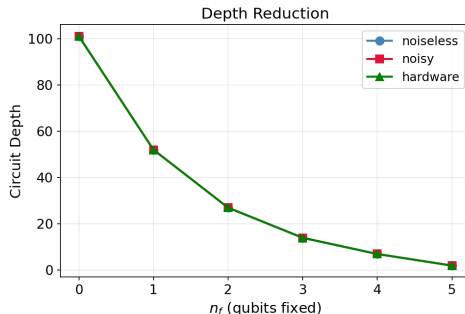
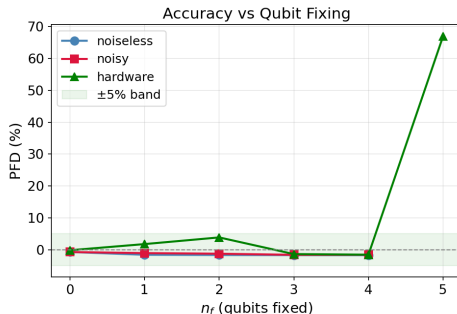
$$[0.03121, 0.05393, 0.04236, 0.03773]$$

QPE output (per-qubit):

$$P_i(1) = [0.096, 0.053, 0.047, 0.044, 0.043]$$

AdaptHHLite Results (H_2)

AdaptHHLite: Noiseless vs Noisy vs Hardware



Key observations:

- Depth reduces dramatically: $101 \rightarrow 52 \rightarrow 27 \rightarrow 14 \rightarrow 7$
- Depth compression: $\frac{D_0 - D}{D_0} \times 100 \Rightarrow 93\%$ at $n_f = 4$
- Accuracy stable up to $n_f = 4$: $|\text{PFD}| \lesssim 2\%$
- Over-fixing ($n_f = 5$) $\Rightarrow$ failure (no valid shots)

NISQ Feasibility

NISQ (Noisy Intermediate-Scale Quantum) devices have limited qubits, noise, and strong constraints on circuit depth.

- Standard HHL requires deep circuits due to quantum phase estimation and controlled unitaries, making it impractical on such hardware.
- HHLite reduces circuit depth through **multiqubit fixing**, while AdaptHHLite **rescales the matrix** to simplify controlled rotations.
- These modifications achieve $\sim 90\%$ **depth compression**, making the algorithm implementable on current devices.
- The authors demonstrate feasibility on real NISQ hardware, including **trapped-ion platforms such as IonQ**
- This comes at the cost of a **small loss in accuracy** ($\sim 1\text{--}2\%$), and excessive fixing can lead to failure.